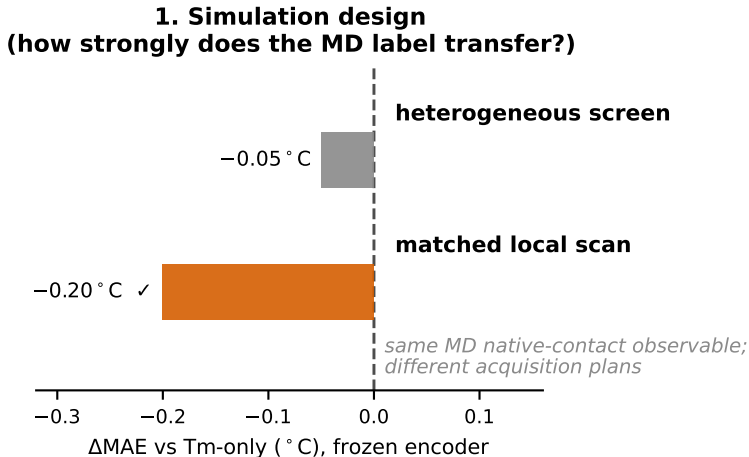


Two choices affect how simulation labels improve low-data nanobody melting-temperature (T_m) prediction



2. Physical observable (does the benefit survive fine-tuning?)

	frozen encoder	fine-tuned (hot)
free energy (FEP)	✓ -0.22 ✓	✓ -0.15 ✓
native-contact Q (MD)	✓ -0.20 ✓	✗ +0.03

Free-energy labels help after encoder fine-tuning; native-contact labels help only with a frozen encoder.